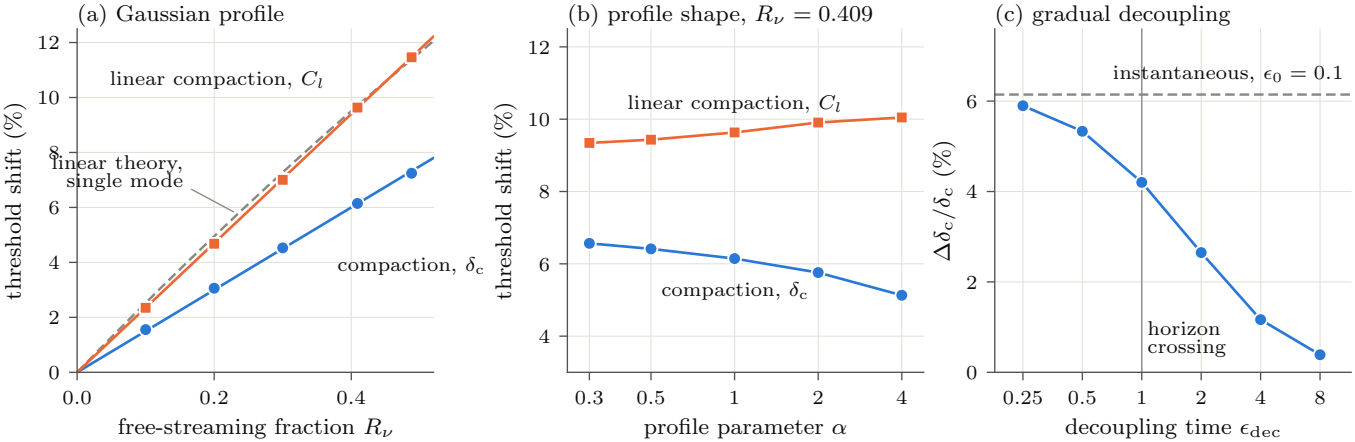

(a) Gaussian profile
linear compaction, C_l
linear theory, single mode
compaction, δ_c
threshold shift (%)
free-streaming fraction R_ν
(b) profile shape, $R_\nu = 0.409$
linear compaction, C_l
compaction, δ_c
threshold shift (%)
profile parameter α
(c) gradual decoupling
instantaneous, $\epsilon_0 = 0.1$
horizon crossing
$\Delta\delta_c/\delta_c$ (%)
decoupling time $\epsilon_{\rm dec}$